

3D Lighter: Learning to Generate Emissive Textures

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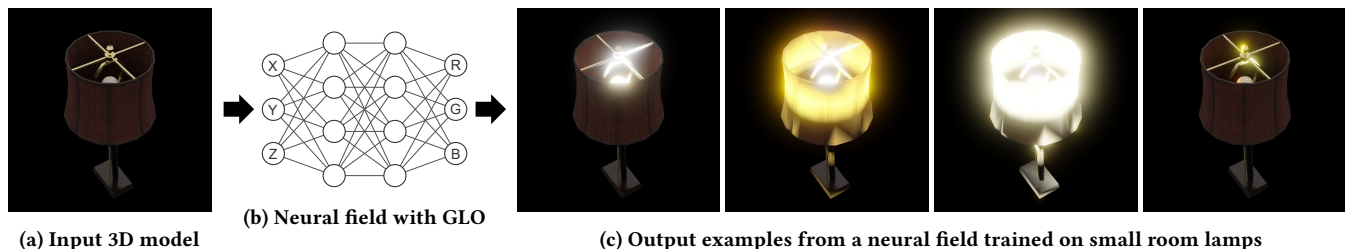


Figure 1: We generate emissive textures for 3D models. Lamps modified from “Abstract Deco Lamp” ©AleixoAlonso (CC BY 4.0).

ABSTRACT

We present a novel approach to generate emissive textures for luminous objects, using direct 3D supervision from a 3D model dataset. To this end, we construct *Emissive Objaverse*, a dataset based on the recently proposed Objaverse dataset, and propose *3D Lighter*, a method using neural fields with generative latent optimization.

ACM Reference Format:

Yosuke Shinya, Kenichi Yoneji, Akihiro Tsukada, and Tatsuya Harada. 2023. 3D Lighter: Learning to Generate Emissive Textures. In *SIGGRAPH Asia 2023 Posters (SA Posters '23)*, December 12–15, 2023, Sydney, NSW, Australia. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3610542.3626153>

1 INTRODUCTION

Luminous objects, such as lamps, greatly affect the scenery of both the real world and games/movies. Emissive textures have become more important to represent luminous objects as physically-based rendering and real-time ray tracing become more popular.

Creating a 3D model that emits light requires a lot of effort because creators must consider which parts emit which colors and draw textures accordingly. To reduce this effort, this paper discusses the automatic generation of emissive textures using machine learning (Figure 1). There are two challenges in generating emissive textures: (1) There is no suitable dataset. Most existing studies use

datasets of OBJ files without emissive textures. The recently proposed Objaverse dataset [Deitke et al. 2023] contains many GLB files in glTF format with emissive textures. However, the variations in Objaverse are quite large, making it difficult to generate emissive textures without additional annotation. (2) There is no suitable method or implementation. In the generation of base color textures, promising results have already been shown (e.g., text to 3D models [Chen et al. 2023] and texture generation for input meshes [Kato et al. 2018; Michel et al. 2022; Richardson et al. 2023]). However, these methods are not designed for generating emissive textures, and their implementations cannot even input or output GLB files.

To address the challenges, we make the following contributions:

- (1) We construct *Emissive Objaverse* based on Objaverse [Deitke et al. 2023], a dataset for the generation of emissive textures.
- (2) We propose *3D Lighter*, a method to generate emissive textures. It uses neural fields with Generative Latent Optimization (GLO) [Bojanowski et al. 2018] and enables 3D supervised learning from 3D model datasets. Our experiments demonstrate its effectiveness. To the best of our knowledge, this is the first work to automatically generate emissive textures by learning luminous 3D models.

2 APPROACH

We tackle the problem of inputting non-luminous 3D models without emissive textures and outputting luminous 3D models with emissive textures. We propose a dataset and a method for solving this problem through 3D supervised learning, which directly reflects the expression and knowledge of creators.

2.1 Dataset

To prepare pairs of non-luminous and luminous 3D models needed for 3D supervised learning, we adopt an approach similar to colorizing grayscale images [Iizuka et al. 2016]. Specifically, we use

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SA Posters '23, December 12–15, 2023, Sydney, NSW, Australia

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ACM ISBN 979-8-4007-0313-3/23/12.

<https://doi.org/10.1145/3610542.3626153>

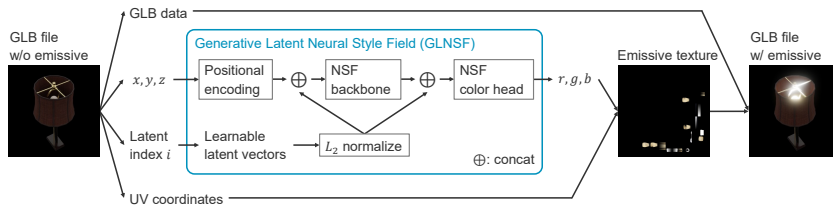
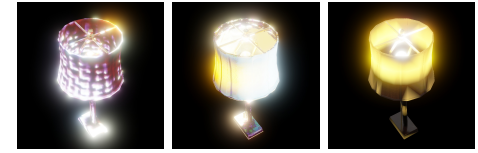


Figure 2: Proposed method. Original lamp: ©AleixoAlonso (CC BY 4.0).



(a) Text2MeshE (b) TEXTureE (c) 3D Lighter

Figure 3: Comparison with other methods. Original lamp: ©AleixoAlonso (CC BY 4.0).

3D models with emissive textures for outputs and use them by dropping emissive textures for inputs.

We downloaded all the 3D models from Objaverse (799 k models) and constructed Emissive Objaverse (75 k models) by extracting ones that have at least one of emissive texture or emissive factor. For the first release, we have annotated lamps (788 models).

2.2 Method

Figure 2 shows the method pipeline with the following three parts.

2.2.1 Inputting GLB and preprocessing. For simplification and acceleration, we unified mesh(es) in each GLB file into one mesh, considering textures (emissive, base color, metallic, roughness, normal) and factor values (e.g., emissive factor) in the glTF format.

2.2.2 Generative Latent Neural Style Field (GLNSF). We use neural fields with Generative Latent Optimization (GLO) [Bojanowski et al. 2018]. The architecture of neural fields is based on Neural Style Field (NSF) of Text2Mesh [Michel et al. 2022]. We use it as a simple color field network by removing layers for displacement.

NSF requires training different neural fields for each output, which is impractical and time-consuming. To solve this problem, we introduce simplified GLO to NSF. We apply L_2 normalization to each latent vector and concatenate it with positional encoding. For the sampling of latent vector indices for GLO, the indices of corresponding training samples are used for training. For inference, we randomly sample them from the training sample indices. Improving the sampling is out of scope of this paper.

Unlike base color textures, emissive textures are often imbalanced with many black areas, making the initial phase of training unstable. In our preliminary experiments, output emissive textures were completely black. We resolved this issue by the bias initialization of RetinaNet [Lin et al. 2017] with $\pi = 0.01$.

Another issue with Text2Mesh is that the output is rough because it optimizes the color of vertices. We improved the sampling of points and enabled the use of UV textures. During training, we randomly sample points on the faces of the mesh (with 50 k points in our experiments).

2.2.3 Outputting GLB and postprocessing. We generate a new emissive texture based on the color outputs of GLNSF and the UV coordinates of the input GLB file. We then export a new GLB file by adding the emissive texture to the input GLB file.

3 EXPERIMENTS

Since there is no existing method to generate emissive textures, for comparison experiments, we modified methods for generating

base color textures so that they can generate emissive textures. We selected recent two methods, Text2Mesh [Michel et al. 2022] and TEXTure [Richardson et al. 2023], and call modified versions Text2MeshE and TEXTureE, respectively. For the two methods, we used a text prompt “an image of a lamp on a black pole” to make non-luminous parts black, partly inspired by 3D Highlighter [Decatur et al. 2023]. For our method, we used a mean squared error (MSE) loss and the Adam optimizer with a learning rate of 10^{-3} , and the latent size of GLO is 64. For a dataset, we used small room lamps (eight models for training and six models for testing).

As shown in Figure 1c, our method can generate various outputs using a single neural field. It has been successful in learning the styles of creators; some set emissive textures only on light bulbs, while others set them also on lampshades. As a failure case, the round metal decoration in the rightmost image is emitting light.

Figure 3 shows the comparison results between the methods. The output of Text2MeshE is rough, and that of TEXTureE is inconsistent depending on viewpoints. Both methods often color the non-luminous parts, which is not suitable for emissive texture generation. We conjecture that the cause of this failure is the large domain gap between web images and emissive textures.

4 CONCLUSION AND LIMITATIONS

We proposed a dataset and a method to generate emissive textures. As shown in the results, our method has insufficient understanding of semantics. Moreover, it cannot learn various categories with a single model. We need to improve the method, and larger/denser datasets and larger models may be necessary to improve practicality and share knowledge between categories.

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